Comprehensive Mixed Conversions Master Set

10-Problem Real-World Applications: Fractions, Decimals, & Percents

Worksheet #105
10 Problem Practice

Student Name: ---------- **Date:** ---------- **Score:** ----------

Instructions: Solve each 10-problem scenario. Convert values across fractions, decimals, and percents as needed, but provide each final response strictly in the **REQUIRED FORMAT**.

PROBLEM 1: AEROSPACE MANUFACTURING — **Format: SIMPLIFIED FRACTION**

1. Aircraft Alloy Composition

An experimental titanium alloy designed for space exploration was synthesized using four primary metal components:

Component Metal	Element Symbol	Mass Share
Titanium	Ti	0.55
Aluminum	Al	3/10
Vanadium	V	12.5%
Iron (Trace)	Fe	*Remaining share*

Calculate the share of Iron (Fe) in the alloy. Express your answer as a **simplified fraction**.

Final Answer (Simplified Fraction): 1/40

PROBLEM 2: MARINE BIOLOGY — **Format: PERCENTAGE (%)**

2. Coral Reef Species Distribution

Marine researchers conducted a biodiversity survey across a 500-square-meter section of a protected coral reef system:

REEF SPECIES PROPORTION

7/20	0.275	20%	?

Hard Corals (7/20) · Soft Corals (0.275) · Sea Anemones (20%) · Sponges (Rest)

What **percentage** of the surveyed reef area is populated by **Sponges**?

Final Answer (Percentage %):

PROBLEM 3: SOFTWARE ENGINEERING — **Format: DECIMAL**

3. Mobile App Storage Usage

A smartphone operating system evaluated the memory consumption of a newly installed social media application:

- **User Data:** 0.38 of total space
- **Cached Media:** 2/5 of total space
- **Executable Code:** 14% of total space
- **Logs & Temp:** Remaining space

Calculate the share of storage taken up by **Logs & Temp files**. Express as a **decimal**.

Final Answer (Decimal):

PROBLEM 4: RENEWABLE ENERGY Format: WHOLE NUMBER (KILOWATT-HOURS)

4. Solar Microgrid Power Output

A community microgrid generated **3,600 Kilowatt-hours (kWh)** of clean solar energy during a summer day:

Sector Destination	Allocation Share	Format Type
Residential Neighborhoods	9/16	Fractional
Commercial Buildings	0.2875	Decimal
Municipal Streetlights	8.75%	Percentage
Battery Reserve Storage	*Remainder*	Unknown Share

.5625
.2875
+ .0875
.9375

Determine the exact energy sent to **Battery Reserve Storage** in whole **Kilowatt-hours (kWh)**.

1 − .9375 = .0625

6.25%

Final Answer (Whole kWh):

PROBLEM 5: FINANCIAL INVESTMENTS Format: DOLLARS & CENTS ($)

5. Portfolio Asset Allocation

An investment manager oversees an institutional portfolio total value of **$240,000** divided across four asset classes:

- **Index Funds:** 0.475 of total portfolio
- **Corporate Bonds:** 1/3 of total portfolio
- **Real Estate Investment Trusts (REITs):** 12.5% of total portfolio
- **Cash Reserves:** The remaining balance

Calculate the exact dollar value held in **Cash Reserves**.

Final Answer (Dollar Amount):

PROBLEM 6: AGRICULTURE & FARMING Format: SIMPLIFIED FRACTION

6. Agricultural Land Utilization

A 1,200-acre commercial farm partitioned its acreage for seasonal crop cultivation:

CROP HARVEST BREAKDOWN

- 🌾 **Corn:** 35% of total acreage
- 🌱 **Soybeans:** 0.325 of total acreage
- 🌾 **Wheat:** 1/5 of total acreage
- 🌻 **Sunflowers:** Remaining acreage

What proportion of the total farm acreage is dedicated to growing **Sunflowers**? Express as a **simplified fraction**.

Final Answer (Simplified Fraction):

PROBLEM 7: TRANSPORTATION & TRANSIT

Format: PERCENTAGE (%)

7. Commuter Mode Split

A metropolitan transit authority surveyed 10,000 daily commuters regarding their primary transport mode:

- **Subway / Rail:** 0.46 of commuters
- **Bus Transit:** 27.5% of commuters
- **Personal Bicycle / Walking:** 7/40 of commuters
- **Rideshare / Personal Vehicles:** The remaining commuters

What **percentage** of commuters use **Rideshare / Personal Vehicles**?

Final Answer (Percentage %):

PROBLEM 8: PUBLIC HEALTH

Format: DECIMAL

8. Nutritional Caloric Breakdown

A nutritionist evaluated the daily macronutrient energy distribution for an athlete consuming 2,800 total calories:

- **Complex Carbohydrates:** 11/20 of total calories
- **Lean Proteins:** 27.5% of total calories
- **Healthy Fats:** 0.14 of total calories
- **Dietary Fiber & Micronutrients:** Remaining calories

Find the share of total calories supplied by **Dietary Fiber & Micronutrients**. Express as a **decimal**.

Final Answer (Decimal):

PROBLEM 9: RETAIL & COMMERCE

Format: WHOLE NUMBER (UNITS)

9. Warehouse Inventory Audit

A fulfillment center holds **4,800 units** of electronics in stock prior to a major holiday sale event:

- **Smartphones:** 3/8 of total inventory
- **Tablets:** 0.2875 of total inventory
- **Laptops:** 22.5% of total inventory
- **Smartwatches:** Remaining inventory

How many individual **Smartwatches** are currently in stock?

Final Answer (Whole Number Units):

PROBLEM 10: ENVIRONMENTAL SCIENCE

Format: DOLLARS & CENTS ($)

10. Ecosystem Grant Distribution

An environmental defense fund awarded **$500,000** in research grants across four conservation projects:

- **Reforestation Initiative:** 0.385 of total grant funding
- **Ocean Debris Cleanup:** 31% of total grant funding
- **Wetland Restoration:** 9/40 of total grant funding
- **Wildlife Sanctuary Expansion:** Remaining funds

Determine the exact funding awarded to the **Wildlife Sanctuary Expansion** project.

Final Answer (Dollar Amount):

Comprehensive Solution Key (Problems 1–10)

Step-by-Step Derivations and Verification

1. Aircraft Alloy (Target: Simplified Fraction)

Ti (0.55) + Al (3/10 = 0.30) + V (12.5% = 0.125) = 0.975. Fe share = 1 - 0.975 = 0.025 = 25/1000 = **1/40**.

2. Coral Reef Species (Target: Percentage %)

Hard Coral (7/20 = 35%) + Soft Coral (0.275 = 27.5%) + Anemones (20%) = 82.5%. Sponges = 100% - 82.5% = **17.5%**.

3. Mobile App Storage (Target: Decimal)

User Data (0.38) + Cache (2/5 = 0.40) + Code (14% = 0.14) = 0.92. Logs & Temp = 1.00 - 0.92 = **0.08**.

4. Solar Microgrid Output (Target: Whole kWh)

Residential (9/16 = 0.5625) + Commercial (0.2875) + Lights (8.75% = 0.0875) = 0.9375. Reserve = 1 - 0.9375 = 0.0625. 3,600 × 0.0625 = **225 kWh**.

5. Portfolio Allocation (Target: Dollar Amount $)

Index (0.475 = 47.5%) + Bonds (1/3 = 33.333%) + REITs (12.5%) = 93.333% (or 112/120 = 14/15). Cash share = 1 - (19/40 + 1/3 + 1/8) = 1 - 112/120 = 8/120 = 1/15. $240,000 × (1/15) = **$16,000**.

6. Agricultural Land (Target: Simplified Fraction)

Corn (35% = 0.35) + Soybeans (0.325) + Wheat (1/5 = 0.20) = 0.875. Sunflowers = 1 - 0.875 = 0.125 = 125/1000 = **1/8**.

7. Commuter Mode Split (Target: Percentage %)

Rail (0.46 = 46%) + Bus (27.5%) + Bike/Walk (7/40 = 17.5%) = 91%. Rideshare = 100% - 91% = **9%**.

8. Nutritional Caloric Breakdown (Target: Decimal)

Carbs (11/20 = 0.55) + Protein (27.5% = 0.275) + Fats (0.14) = 0.965. Micronutrients = 1.000 - 0.965 = **0.035**.

9. Warehouse Inventory (Target: Whole Units)

Smartphones (3/8 = 0.375) + Tablets (0.2875) + Laptops (22.5% = 0.225) = 0.8875. Smartwatches = 1 - 0.8875 = 0.1125. 4,800 × 0.1125 = **540 units**.

10. Ecosystem Grant Distribution (Target: Dollar Amount $)

Reforestation (0.385 = 38.5%) + Cleanup (31%) + Wetland (9/40 = 22.5%) = 92%. Sanctuary = 100% - 92% = 8% (0.08). $500,000 × 0.08 = **$40,000**.